Math 2059 Homework 2

Deadline: 2pm, 25th December 2025.

Preparation and submission: You can discuss about the solutions with your friends, but you must solve the questions on physical papers with your own handwriting. You must **submit** the homework to me **in person,** either during the lectures or by handing it to me when you find me in my office (M2 232). The HW sent via online channels such as e-mail will be ignored and won't be considered for grading!

Note: The answers in this HW may not necessarily be Integers. For consistency and precision, please present your results using **three digits after the decimal point** if the result is not Integer.

Problem 1) A number of matrices are defined as:

$$[A] = \begin{bmatrix} 4 & 7 \\ 1 & 2 \\ 5 & 6 \end{bmatrix} \qquad [B] = \begin{bmatrix} 4 & 3 & 7 \\ 1 & 2 & 7 \\ 2 & 0 & 4 \end{bmatrix}$$

$$\{C\} = \begin{Bmatrix} 3 \\ 6 \\ 1 \end{Bmatrix} \qquad [D] = \begin{bmatrix} 9 & 4 & 3 & -6 \\ 2 & -1 & 7 & 5 \end{bmatrix}$$

$$[E] = \begin{bmatrix} 1 & 5 & 8 \\ 7 & 2 & 3 \\ 4 & 0 & 6 \end{bmatrix}$$

$$[F] = \begin{bmatrix} 3 & 0 & 1 \\ 1 & 7 & 3 \end{bmatrix} \qquad \lfloor G \rfloor = \lfloor 7 \quad 6 \quad 4 \rfloor$$

Perform the following operations, where **applicable**. Show all steps of the computation.

(1) [E] + [B]
(2) [A] × [F]
(3) [B] − [E]
(4) 7 × [B]
(5) [E] × [B]
(6) $\{C\}^T$
(7) [B] × [A]
(8) $[D]^T$
(9) [A] × {C}
(10) [I] × [B]
(11) $[E]^T[E]$
(12) $\{C\}^T\{C\}$

Problem 2) For the set of equations:

$$2x_2 + 5x_3 = 9$$

$$2x_1 + x_2 + x_3 = 9$$

$$3x_1 + x_2 = 10$$

(a) Compute the determinant.
(b) Use **Cramer's rule** to solve for the x's. Show all steps of the computation.
(c) Substitute your results back into the original equation to check your results.

Problem 3) Given the equations:

$$10x_1 + 2x_2 - x_3 = 27$$

$$-3x_1 - 6x_2 + 2x_3 = -61.5$$

$$x_1 + x_2 + 5x_3 = -21.5$$

(a) Solve by **naive Gauss elimination**. Show all steps of the computation.
(b) Substitute your results into the original equations to check your answers.

Problem 4) Use **Gauss-Jordan** elimination to solve:

$$2x_1 + x_2 - x_3 = 1$$

$$5x_1 + 2x_2 + 2x_3 = -4$$

$$3x_1 + x_2 + x_3 = 5$$

Problem 5) Use **LU decomposition** to determine the matrix inverse for the following system. Show all steps of the computation. Do not use a pivoting strategy, and check your results by verifying that $[A][A]^{-1} = [I]$.

$$10x_1 + 2x_2 - x_3 = 27$$

$$-3x_1 - 6x_2 + 2x_3 = -61.5$$

$$x_1 + x_2 + 5x_3 = -21.5$$

Problem 6) Use the **Gauss-Seidel** method to solve the following system until the percent relative error falls below $\varepsilon_s = 5\%$. Show all steps of the computation.

$$10x_1 + 2x_2 - x_3 = 27$$

$$-3x_1 - 6x_2 + 2x_3 = -61.5$$

$$x_1 + x_2 + 5x_3 = -21.5$$